GF/PED Manual

Wilson GF analysis, internal-coordinate force constants and potential-energy distributions in ORACLE

ORACLE Development Notes

June 27, 2026

Abstract

`GF/PED` is the `ORACLE` workflow that transforms a Cartesian harmonic Hessian into an internal-coordinate Wilson GF problem, solves the harmonic vibrational eigenproblem and reports frequencies, internal force constants, normal modes and potential-energy distributions (PEDs). The production path uses a frozen non-redundant `GICForge` definition, so vibrational analysis, Gaussian input generation and structural refinement can share the same coordinate model. This manual describes the theory, input contracts, command line and GUI workflows, Pulay-style scaling, output files and validation rules.

Contents

1 Purpose

The GF/PED module answers a specific question: given a Cartesian Hessian and a non-redundant internal-coordinate basis, what are the harmonic frequencies, normal modes, internal force constants and coordinate contributions in that basis? It is deliberately separate from VPT2/VCI. VPT2/VCI works in normal-coordinate quartic force fields; GF/PED works with a Cartesian Hessian, a Wilson B matrix and a non-redundant coordinate model.

The preferred production workflow is `gic-gf`. It reads a frozen `GICForge` schema and a Gaussian FCHK Hessian adapter, evaluates the same GICs on the Hessian geometry or on an optional external geometry, transforms the Cartesian Hessian to the GIC basis and solves Wilson GF. A simpler diagnostic workflow, `gf`, reads an FCHK file and generates a ORACLE GIC basis on the fly; it is useful for quick checks but does not provide the same cross-module coordinate identity as a frozen `GICForge` schema.

2 Theoretical Background

Let x be the $3N$-dimensional Cartesian coordinate vector, F_x the Cartesian harmonic Hessian in E_h/bohr^2, and m_i the atomic masses in amu. For a non-redundant internal coordinate vector q, the linearized relation between Cartesian and internal displacements is

$$dq = B\,dx,$$

where $B = \partial q/\partial x$ is the Wilson B matrix. The kinetic-energy matrix in internal coordinates is

$$G = B\,M^{-1}\,B^T,$$

where M^{-1} is the diagonal Cartesian inverse-mass matrix with each atomic mass repeated for x, y, z.

The internal-coordinate Cartesian backtransform used by the code is

$$A = M^{-1}B^TG^+,$$

where G^+ is the numerical inverse or pseudoinverse of G. The internal force-constant matrix is then

$$F_q = A^TF_xA.$$

The matrix is explicitly symmetrized before the GF solution. If Pulay-style diagonal scaling factors s_i are supplied, the scaled internal Hessian is

$$(F_q^{\text{scaled}})_{ij} = (F_q)_{ij}\sqrt{s_is_j}.$$

This preserves symmetry and applies off-diagonal scaling as the geometric mean of the two diagonal factors.

The Wilson GF eigenproblem is solved in symmetric form. Since G is positive definite for a valid non-redundant basis, the code diagonalizes

$$G^{1/2}F_qG^{1/2}.$$

The eigenvalues are converted to signed harmonic frequencies in cm^{-1}. Negative eigenvalues are retained as negative frequencies and therefore expose transition-state or instability directions rather than being silently clipped.

The internal normal modes are back-expressed in the GIC basis, and the potential-energy distribution for coordinate i in mode k is evaluated from the mode vector l_k as

$$\mathrm{PED}_{ik} = 100\,\frac{|l_{ik}(F_q l_k)_i|}{\sum_j |l_{jk}(F_q l_k)_j|}.$$

The absolute-value normalization makes the PED table robust for coupled modes with positive and negative cross terms, while retaining the relative coordinate contribution pattern.

3 Input Contracts

3.1 Cartesian Hessian

The canonical internal object is `oracle_gf.HessianInput`. It stores atomic numbers, Cartesian coordinates in bohr, atomic masses in amu, a full symmetric Cartesian Hessian in E_h/bohr^2, optional harmonic frequencies from the source adapter and provenance metadata. The current file adapter is Gaussian FCHK/FCH.

The FCHK file must contain:

- atomic numbers;
- current Cartesian coordinates;
- atomic masses;
- Cartesian force constants;
- harmonic frequencies when available.

GF/PED does not read geometries from Gaussian log text and does not use a Z-matrix. The FCHK adapter is only a source of canonical Cartesian Hessian data.

3.2 Frozen GIC Definition

The production branch reads a `oracle.gic.definition.v1` JSON schema created by `gic-define`. The schema contains primitive coordinates, the GIC coefficient matrix, labels, names, irreducible representations, point group, symmetrization flag and the Gaussian-readable GIC block. GF/PED evaluates that frozen definition; it does not rerun topology perception, pruning or symmetry assignment.

The atom count in the schema and in the FCHK input must agree. If an optional geometry is supplied for the B matrix, it must describe the same atom ordering and the same topology. Coordinates supplied through `-geometry` are in angstrom and are converted internally to bohr before B-matrix evaluation.

4 Command-Line Workflows

4.1 Quick FCHK workflow

The quick workflow builds a ORACLE GIC basis directly from the FCHK geometry:

```
python -m matrix gf \
  --fchk working/gauin.fchk \
  --out working/gf_ped_report.txt \
  --csv-dir working/gf_csv
```

This branch is useful for diagnostics. Because the GICs are generated inside the GF run, it should not be used when the analysis must be identical to a previous Gaussian input, SEfit/MORPHEUS refinement or GICForge manifest.

4.2 Frozen-GIC production workflow

The production workflow first freezes the coordinate model and then runs GF/PED:

```
python -m matrix gic-define \
  --geometry parent.xyz \
  --out working/gic_definition.json \
  --gaussian-out working/gauin

python -m matrix gic-gf \
  --schema working/gic_definition.json \
  --fchk working/gauin.fchk \
  --out working/gic_gf_ped_report.txt \
  --csv-dir working/gic_gf_csv
```

If the B matrix must be evaluated on a geometry different from the FCHK geometry, pass:

```
python -m matrix gic-gf \
  --schema working/gic_definition.json \
  --fchk working/gauin.fchk \
  --geometry working/current.xyz \
  --out working/gic_gf_ped_report.txt
```

This is useful for controlled diagnostics, but production vibrational analysis should normally evaluate the B matrix on the same equilibrium geometry used for the Hessian.

5 Pulay-Style Scaling

GF/PED accepts optional diagonal scaling factors in the GIC basis. Scaling is applied after transformation of the Cartesian Hessian to internal coordinates and before the Wilson GF eigenproblem. A scaling file is line-oriented text or CSV:

```
# selector factor
default 1.000
GIC003 0.980
A1Str0001,0.995
5=1.020
class CH_stretches 0.970 R(1,6)|R(2,7)|R(3,8)
```

Selectors may be:

- `default`, `all` or `*`;
- a one-based index such as `5`;

- a `GICnnn` label;
- an exact GIC name;
- an exact label;
- an unambiguous substring of a name or label.

Inline records are supplied with repeated `-scale` options:

```
python -m matrix gf \
  --xyzin working/molecule.xyzin \
  --fchk working/gauin.fchk \
  --scale-file working/pulay_scale.txt \
  --scale "GIC003=0.980" \
  --scale "default=1.000" \
  --scale-class "CH_stretches:0.970:R(1,6)|R(2,7)|R(3,8)"
```

Class records reuse the semiexperimental-fit idea of named parameter classes, but apply it to diagonal Pulay factors in the frozen GIC basis. A class has a name, one factor and one or more case-insensitive label/name patterns separated by `|`. Patterns in a class may deliberately match several GICs. MATRIX rejects the class if it matches no GIC or if the matched GICs mix coordinate families such as stretches, bends, torsions, out-of-plane coordinates, ring puckering coordinates or special fragment/center coordinates. This prevents a chemically motivated scale factor from silently spanning unrelated internal coordinate types. Scaling factors must be non-negative. Ambiguous or unmatched ordinary selectors remain fatal errors because silent scaling of the wrong coordinate would invalidate the PED interpretation.

6 Non-Bonded Hessian Subtraction and Local Models

Before the Cartesian Hessian is transformed to the GIC basis, GF/PED may subtract model electrostatic and van der Waals contributions:

```
python -m matrix gf \
  --fchk working/gauin.fchk \
  --xyzin working/xyzin \
  --subtract-electrostatic \
  --subtract-uff-vdw \
  --nonbonded-14-scale 0.5 \
  --local
```

The charge vector is read from the canonical `#SYNTHONS` section. If Gaussian CM5 charges were imported during LINK preprocessing, those are used. Otherwise the same synthon charge column is present and comes from the ORACLE electronegativity model. GF/PED therefore never reparses Gaussian text for charges.

The van der Waals correction uses ORACLE topology radii and UFF well-depth coefficients where available. Unsupported element pairs fail explicitly rather than silently using zeros.

Both electrostatic and UFF-vdW corrections use the same topological pair policy: 1-2 and 1-3 pairs are excluded, 1-4 pairs are included with `-nonbonded-14-scale` (default 0.5), and pairs beyond 1-4 are included with full weight. Disconnected inter-fragment pairs are treated as beyond 1-4.

The optional `-local` model is applied after this Cartesian subtraction and after transformation to internal coordinates. It keeps diagonal primitive terms and local bonded couplings while zeroing non-local internal force-constant terms.

7 Output Files

Output	Meaning
`gf_ped_report.txt`	Human-readable report for the quick `gf` workflow.
`gic_gf_ped_report.txt`	Human-readable report for the frozen-GIC production workflow.
`gf_manifest.json`	Run manifest for the quick workflow.
`gic_gf_manifest.json`	Run manifest recording FCHK, schema, optional geometry, scaling input and output files.
`*_frequencies.csv`	Mode index and frequency in cm^{-1}.
`*_gic_labels.csv`	GIC index, name, irrep and coordinate label.
`*_ped.csv`	Potential-energy distribution, rows as GICs and columns as modes.
`*_normal_modes.csv`	Internal-coordinate normal-mode vectors.
`*_force_constants.csv`	Internal-coordinate force-constant matrix after optional scaling.
`*_g_matrix.csv`	Wilson G matrix in the same GIC order.

The text report begins with the source FCHK, coordinate source, point group, symmetrization flag and GIC count. It then prints frequencies, GIC labels and the PED matrix. In the frozen-GIC workflow, labels and irreps are those stored or evaluated from the `GICForge` schema.

8 GUI Workflow

The advanced GUI exposes GF/PED through the dedicated `GF / PED` window. The window reads a Cartesian Hessian FCHK/FCH file, an optional frozen GIC definition, an optional B-matrix geometry and optional Pulay scaling records. It displays the text report and can export the same CSV tables written by the CLI.

The dashboard also exposes a `GIC Definition / B Matrix / GF-PED` workflow. The intended sequence is:

1. define or load a frozen GIC schema;
2. select the FCHK Hessian;
3. optionally select a current geometry for B evaluation;
4. optionally supply Pulay scaling;
5. run GF/PED and export report/CSV tables.

9 Validation and Failure Policy

GF/PED must fail rather than continue when the mathematical model is invalid. Fatal conditions include:

- FCHK missing masses, Cartesian coordinates or Cartesian force constants;
- non-symmetric Cartesian Hessian;
- schema atom count different from the Hessian input atom count;
- B matrix dimension inconsistent with the Hessian;
- non-positive-definite Wilson G matrix for the selected coordinates;
- negative Pulay scaling factor;
- ambiguous or unmatched Pulay selector.

A negative vibrational eigenvalue is not a fatal error by itself. It is reported as a negative frequency because it may represent a transition-state direction or an instability in the supplied Hessian/geometry combination.

10 Relation to GICForge, MORPHEUS and VPT2/VCI

`GICForge` owns coordinate construction, pruning and symmetry labels. GF/PED consumes that frozen coordinate model and performs harmonic vibrational linear algebra. MORPHEUS/SEfit consumes the same coordinate model for semiexperimental structure refinement. VPT2/VCI is a separate workflow based on normal-coordinate cubic and quartic force fields and should not be confused with GIC-based Wilson GF/PED.

This separation is intentional. A coordinate definition can be generated once, used to prepare Gaussian input, reused for semiexperimental refinement, and then reused again for GF/PED analysis of the Cartesian Hessian. When the same schema is used throughout, coordinate labels, symmetry species, B matrices and PED assignments remain auditable across modules.